David Hany

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EDUCATION

Pharos University in Alexandria (PUA) **2022 – Expected 2026**

B.Sc. in Computer Science and Artificial Intelligence (Cyber Security Specialization) — GPA 2.7 Alexandria, Egypt

EXPERIENCE

WE Telecommunications **2023**

Cyber Security Intern (120 Hours Intensive Practical Program) Alexandria, Egypt

- Conducted targeted vulnerability assessments and penetration testing simulations across internal telecommunications networks utilizing **Metasploit** and Kali Linux.
- Captured and inspected live network packets with **Wireshark** to proactively identify potential reconnaissance attempts, protocol anomalies, and unencrypted traffic.
- Assisted senior security analysts in designing, evaluating, and implementing proactive network defense strategies and perimeter firewall policies.

PROJECTS & OPEN SOURCE (GITHUB REPOSITORIES)

UpHeal / MindQuest (Graduation Project) | *Android Security, Edge AI, RASP* `github.com/davidhany73 ↗`

- Engineered a multi-layered Android security paradigm integrating Host-Based Intrusion Prevention (HIPS), Edge AI for real-time visual classification, and Runtime Application Self-Protection (RASP) against reverse engineering and dynamic binary tampering.

Fraud Bank Detection System | *Python, Machine Learning, Anomaly Detection, Scikit-Learn* `Fraud-Bank-Detection ↗`

- Architected an automated banking fraud detection pipeline using machine learning (Scikit-Learn, Pandas) to analyze high-volume transaction telemetry, detect anomalies in real-time, and flag unauthorized transfers with minimal false positives.

Multimodal Biometric Attendance System | *Python, OpenCV, Computer Vision, Facial Recognition* `Biometric-Attendance ↗`

- Developed an autonomous biometric verification platform featuring real-time facial recognition and multimodal feature matching to eliminate proxy sign-ins, ensure strict identity validation, and log timestamped records.

Encrypted Secure Chat Application | *C, Cryptography, OpenSSL, Socket Programming* `Encrypted-Secure-Chat ↗`

- Built a high-performance low-level client-server messaging system in C featuring end-to-end cryptographic encryption, secure key exchange mechanisms, and hardened socket communication to prevent eavesdropping and Man-in-the-Middle (MitM) attacks.

Ethereum Web3 Crypto Wallet | *JavaScript, Web3.js, Ethereum, Smart Contracts* `Ethereum-Web3-Wallet ↗`

- Implemented a decentralized Web3 Ethereum wallet application supporting client-side private key generation, transaction signing, balance monitoring, and automated smart contract interaction on Ethereum networks.

PROJECTS & OPEN SOURCE (CONTINUED)

Brute-Force & C2 Attack Simulator | *Python, Cryptanalysis, Red Teaming, Multithreading* `Brute-Force-Simulator ↗`

- Created educational offensive security modules simulating multi-threaded credential brute-forcing, dictionary attacks, and Command & Control (C2) payloads to benchmark authentication entropy and evaluate intrusion defense thresholds.

Password Strength & Entropy Analyzer | *Python, Information Theory, NIST Guidelines* `Password-Checker ↗`

- Engineered a security validation utility calculating Shannon entropy, dictionary leak vulnerability, and character variance against NIST SP 800-63B guidelines to enforce resilient credential policies.

Instagram Clone (Flask & MySQL) | *Python, Flask, MySQL, RESTful APIs, Session Security* `Instagram-Clone-Flask ↗`

- Engineered a full-stack relational database application with secure user authentication, password hashing, session tokens, media storage pipelines, and normalized MySQL schemas optimized for query performance.

Smart Glove Robotic Hand Interface | *C++, Arduino, Embedded Systems, IoT Robotics* `Smart-Glove-Robotic ↗`

- Programmed an embedded hardware interface translating flex sensor telemetry via microcontroller ADC channels to actuate an articulated multi-servo robotic hand in real time with minimal latency.

Smart Attendance System (QR Code) | *Python, Web Architecture, Cryptographic QR Tokens* `Smart-Attendance-QR ↗`

- Constructed a dynamic QR attendance verification platform utilizing time-expiring cryptographic tokens and session cookies to mitigate fraudulent check-ins and streamline automated record auditing.

Security Review & Access Governance Engine | *RBAC, Access Control, Security Auditing* `Access-Control-Engine ↗`

- Implemented a security auditing architecture demonstrating Role-Based Access Control (RBAC), privilege boundary separation, and permission matrix evaluation to prevent unauthorized privilege escalation.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, JavaScript (ES6+), SQL, HTML/CSS

Cybersecurity & Tools: Metasploit, Wireshark, Kali Linux, Nmap, Cryptography & PKI, End-to-End Encryption, Socket Programming, Linux System Hardening

AI, Vision & Data: OpenCV, Machine Learning (Scikit-Learn), TensorFlow Lite, Edge AI, NumPy, Pandas, MATLAB

Web, Backend & IoT: React, Node.js, Web3.js, Flask, MySQL Database Architecture, Arduino Embedded C++, RESTful APIs

Professional Competencies: Vulnerability Assessment, Threat Modeling, Red/Blue Team Tactics, Analytical Problem-Solving, Team Collaboration

CERTIFICATIONS & AWARDS

Cisco CCNA (200-301) Training Program (120 Hours) — WE Telecommunications **Aug 2024 – Sep 2024**

CompTIA Security+ Certification — CompTIA Online **2024**

Google Cybersecurity Professional Certificate — Coursera • Google **2023**

Certificate of Appreciation (Talent Day Show) — Pharos University in Alexandria **Feb 2023**

LANGUAGES

Arabic: Native Proficiency • **English:** Professional Working Proficiency